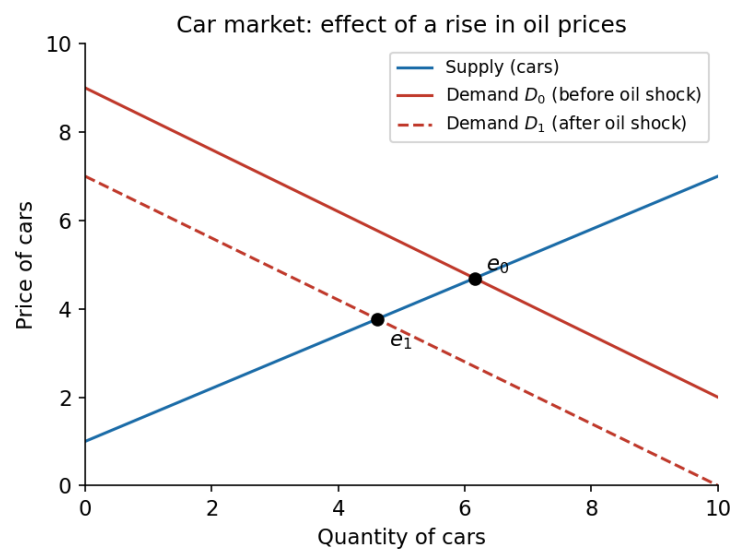


Problem 1

A political conflict decreases global aggregate oil supply, so the oil supply curve shifts left and the equilibrium price of oil rises.

1. Effect on the demand for cars

Gasoline (refined from oil) and cars are complements — you need fuel to operate a car, so the cost of using a car rises when oil prices rise. A higher cost of a complementary good reduces demand for the good itself: the demand curve for cars shifts left ($D_0 \rightarrow D_1$), while the supply of cars is unaffected. The new equilibrium has both a lower price and a lower quantity of cars sold.

**2. Effect on the demand for alternative energies**

Alternative energy sources (solar, wind, electric power, etc.) are substitutes for oil. As oil becomes more expensive, consumers and firms shift toward relatively cheaper substitutes — demand for alternative energy increases (the demand curve shifts right), raising both its equilibrium price and quantity.

Problem 2

Using Fig. 4.2 from the assignment: demand curves $x_1 < x_2 < x_3$ (increasing demand, left to right) and supply curves $y_1 < y_2 < y_3$ (increasing supply, left to right). The current equilibrium is $e = x_2 \cap y_2$. Each part below asks which curve(s) shift and identifies the resulting point on the grid.

1. Trade union enforces a wage increase

Higher wages raise the dairy's production cost, which decreases supply at every price: the supply curve shifts left, from y_2 to y_1 . Demand is unaffected (stays at x_2). New equilibrium: point b ($x_2 \cap y_1$) — higher price, lower quantity than e.

2. Recession reduces disposable income (cheese is a normal good)

Because cheese is a normal good, lower income directly reduces demand at every price: the demand curve shifts left, from x_2 to x_1 . Supply is unaffected (stays at y_2). New equilibrium: point d ($x_1 \cap y_2$) — lower price and lower quantity than e.

3. Study reports cheese improves intellectual abilities

Positive information about a good's benefits raises consumers' willingness to buy it at every price: demand shifts right, from x_2 to x_3 . Supply is unaffected (stays at y_2). New equilibrium: point f ($x_3 \cap y_2$) — higher price and higher quantity than e.

4. Epidemic cuts the dairy-producing livestock in half

Losing half the milk supply is a large negative input shock, which decreases supply at every price: the supply curve shifts left, from y_2 toward y_1 (demand unaffected, stays at x_2). On this three-curve grid the qualitative direction is identical to part 1, landing at point b — higher price, lower quantity than e. In reality, losing half the livestock is a far more severe supply shock than a wage increase, so the actual leftward shift would likely be larger than $y_2 \rightarrow y_1$ suggests (potentially off this grid entirely) — but among the three supply curves given, y_1 is the only one that captures the correct direction of the effect.

Problem 3

1. Individual and aggregate weekly demand

Urs: at $p = 0$, demand = 10 scoops/week; each CHF 1 increase reduces quantity by 2/week $\rightarrow x_U(p) = 10 - 2p$, valid for $0 \leq p \leq 5$.

Heidi: at $p = 0$, demand = 2 scoops/day (=14/week); each CHF 1 increase reduces quantity by 1/day (=7/week) $\rightarrow x_H(p) = 14 - 7p$, valid for $0 \leq p \leq 2$.

p (CHF)	Urs (per week)	Heidi (per week)	Aggregate demand
0	10	14	24
1	8	7	15
2	6	0	6
3	4	0	4
4	2	0	2
5	0	0	0

Market demand function (aggregate, piecewise since Heidi's demand hits 0 at $p = 2$):

$$x(p) = 24 - 9p \quad \text{for } 0 \leq p \leq 2$$

$$x(p) = 10 - 2p \quad \text{for } 2 < p \leq 5$$

$$x(p) = 0 \quad \text{for } p > 5$$

2. Is ice cream an ordinary good for Urs and Heidi?

Yes. Both individual demand functions are downward sloping (quantity demanded falls as price rises), which is the definition of an ordinary good — a Giffen good would instead have an upward-sloping demand curve.

3. iScream's supply

iScream will not supply at $p = 0$ and increases supply by 15 scoops/week for each CHF 1 price increase, so its supply function passes through the origin with slope 15:

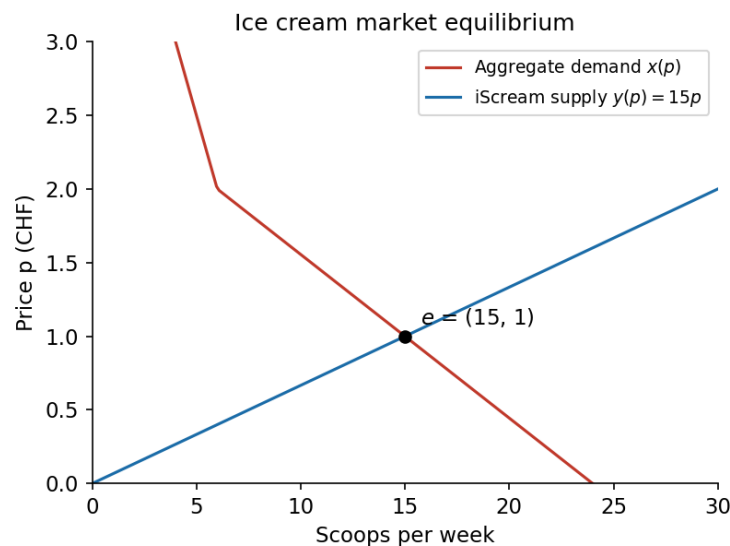
$$y(p) = 15p, \text{ for } p \geq 0$$

p (CHF)	iScream supply (per week)
0	0
1	15
2	30
3	45
4	60
5	75

(a) Supply function — see above: $y(p) = 15p$.

(b) Market equilibrium

For $0 \leq p \leq 2$, set demand = supply: $24 - 9p = 15p \Rightarrow 24 = 24p \Rightarrow p^* = 1$ (within range, so valid).
Equilibrium quantity: $x^* = y^* = 15 \cdot 1 = 15$ scoops/week.



(c) Alexander joins Urs (buys the same amount as Urs)

Urs' contribution effectively doubles. New aggregate demand for $0 \leq p \leq 2$: $2 \cdot (10 - 2p) + (14 - 7p) = 34 - 11p$. Setting this equal to supply: $34 - 11p = 15p \Rightarrow 34 = 26p \Rightarrow p^* = 17/13 \approx \text{CHF } 1.31$.

Equilibrium quantity: $x^* = 15 \cdot (17/13) = 255/13 \approx 19.6$ scoops/week. Both the equilibrium price and quantity rise compared to part (b) ($p = 1, x = 15$), since Alexander's demand shifts the aggregate demand curve to the right.

Problem 4

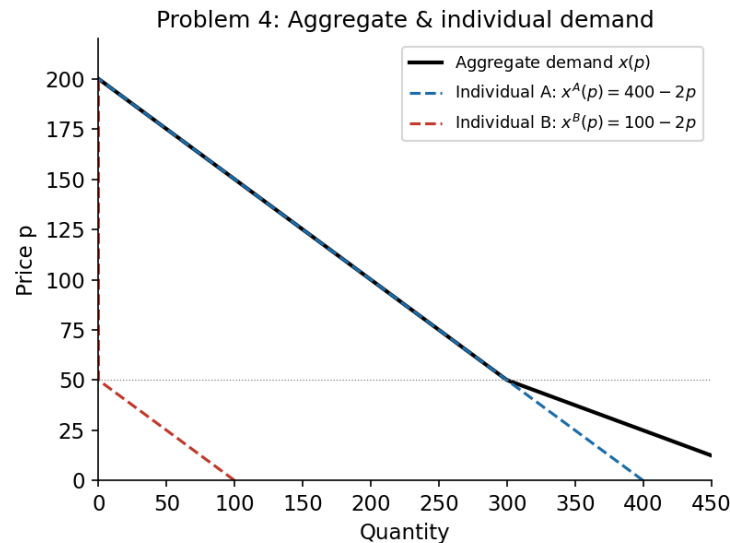
1. Individual demand functions

The aggregate demand kink at $p = 50$ (slope changes from -4 below 50 to -2 at/above 50) means one individual's demand hits zero exactly at $p = 50$, leaving only the other individual active above that price. Solving backward from the two pieces:

$$x^A(p) = 400 - 2p, \text{ for } 0 \leq p \leq 200 \text{ (demand hits 0 at } p = 200)$$

$$x^B(p) = 100 - 2p, \text{ for } 0 \leq p \leq 50 \text{ (demand hits 0 at } p = 50)$$

Check: for $p \geq 50$, only A is active: $x^A(p) = 400 - 2p$ ✓. For $p < 50$, $x^A(p) + x^B(p) = (400 - 2p) + (100 - 2p) = 500 - 4p$ ✓, matching the given aggregate demand.



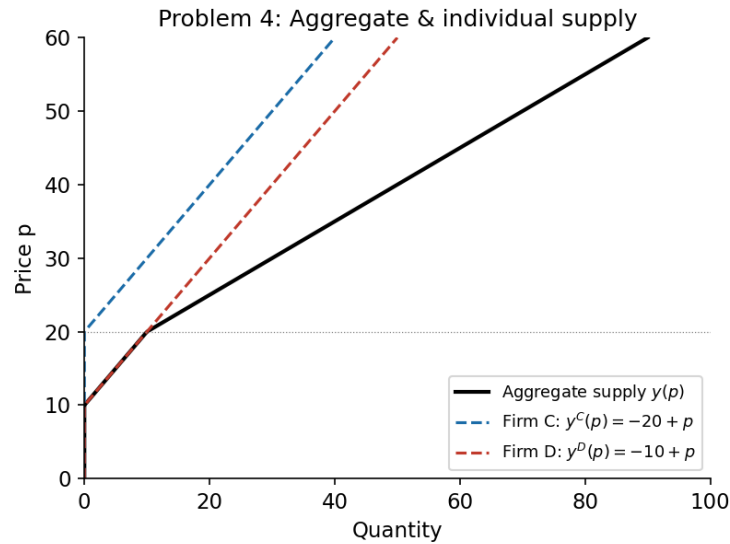
2. Individual supply functions

The aggregate supply kink at $p = 20$ (slope changes from 1 below 20 to 2 at/above 20) means one firm's supply hits zero exactly at $p = 20$:

$$y^C(p) = -20 + p, \text{ for } p \geq 20 \text{ (supply is 0 below } p = 20)$$

$$y^D(p) = -10 + p, \text{ for } p \geq 10 \text{ (supply is 0 below } p = 10)$$

Check: for $10 \leq p < 20$, only D is active: $y^D(p) = -10 + p$ ✓. For $p \geq 20$, $y^C(p) + y^D(p) = (-20 + p) + (-10 + p) = -30 + 2p$ ✓, matching the given aggregate supply.



3. Equilibrium price and quantity

Try the $p \geq 50$ demand piece against the $p \geq 20$ supply piece: $400 - 2p = -30 + 2p \Rightarrow 430 = 4p \Rightarrow p^* = 107.5$, which is indeed ≥ 50 and ≥ 20 , so this is consistent.

$$p^* = 107.5, \quad y^* = -30 + 2(107.5) = 185 \quad (\text{matches } x(107.5) = 400 - 215 = 185 \checkmark)$$

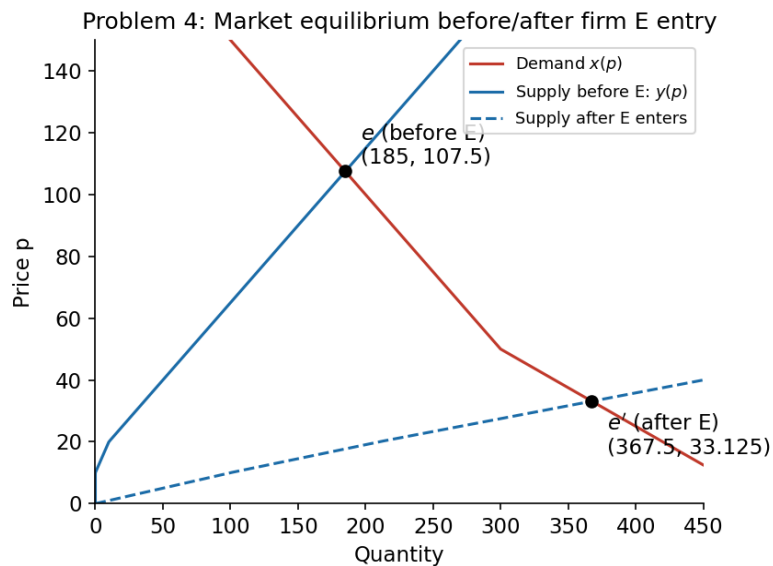
4. Individual equilibrium quantities

At $p^* = 107.5$ (above B's shutdown price of 50), only A is active on the demand side:

- $x^A = 400 - 2(107.5) = 185$
- $x^B = 0$ (B's demand formula gives a negative number at this price, so B simply buys nothing)

On the supply side, both firms are active ($p^* \geq 20$):

- $y^C = -20 + 107.5 = 87.5$
- $y^D = -10 + 107.5 = 97.5$ (sum = 185 $\checkmark$)



5. Does firm E's entry change the results?

Firm E adds supply $y^E(p) = 10p$ (active at any $p \geq 0$). New aggregate supply: for $p \geq 20$, $y(p) = (-20+p) + (-10+p) + 10p = -30 + 12p$. Testing this against the demand piece for $p < 50$ ($500 - 4p$), since the old equilibrium price of 107.5 is no longer likely to hold once a large new supplier enters:

$$500 - 4p = -30 + 12p \Rightarrow 530 = 16p \Rightarrow p^* = 33.125 \quad (\text{confirmed in range } 20 \leq p < 50)$$

$$y^* = -30 + 12(33.125) = 367.5 \quad (\text{matches } x(33.125) = 500 - 132.5 = 367.5 \checkmark)$$

Yes — E's entry substantially changes both results. The equilibrium price falls from 107.5 to 33.125 and quantity rises from 185 to 367.5. At the new, much lower price, individual B re-enters the demand side (since $33.125 < 50$):

- $x^{A*} = 400 - 2(33.125) = 333.75$
- $x^{B*} = 100 - 2(33.125) = 33.75$ (sum = 367.5 $\checkmark$)
- $y^{C*} = -20 + 33.125 = 13.125$
- $y^{D*} = -10 + 33.125 = 23.125$
- $y^{E*} = 10(33.125) = 331.25$ (sum = 367.5 $\checkmark$ — firm E supplies the large majority of the market)

Problem 5

1. Maximum number of goods that can be inferior

At most $n - 1$ of the n goods can be inferior. Sabine spends her entire income b on the n goods, so total expenditure always equals b (the budget constraint binds with equality). If income rises and every single good were inferior, quantity demanded — and hence spending — would fall for all n goods simultaneously, so total expenditure would fall below the new, higher income. That contradicts the requirement that all income is spent. So at least one good must be normal (its demand does not fall when income rises) to absorb the additional spending; the other $n - 1$ goods are free to be inferior.

2. Demand for good 1: $x_1(b, p_1, p_2) = b / (p_1 + a \cdot p_2)$

(a) Substitutes or complements?

Cross-price effect: $\partial x_1 / \partial p_2 = -ab / (p_1 + a \cdot p_2)^2$. Since $b > 0$ and the denominator is squared (always positive), the sign of this derivative is the opposite sign of a .

- $a > 0 \Rightarrow \partial x_1 / \partial p_2 < 0 \Rightarrow$ goods 1 and 2 are complements (a price rise in good 2 lowers demand for good 1).
- $a < 0 \Rightarrow \partial x_1 / \partial p_2 > 0 \Rightarrow$ goods 1 and 2 are substitutes (a price rise in good 2 raises demand for good 1).
- $a = 0 \Rightarrow \partial x_1 / \partial p_2 = 0 \Rightarrow$ the goods are independent — demand for good 1 reduces to $x_1 = b/p_1$ and doesn't respond to p_2 at all.

(b) For $a > 0$: normal or inferior? Ordinary or Giffen?

Income effect: $\partial x_1 / \partial b = 1 / (p_1 + a \cdot p_2) > 0$ (positive, since $p_1 + a \cdot p_2 > 0$ is guaranteed by the stated condition $p_1 > -a \cdot p_2$). Demand rises with income, so good 1 is a normal good.

Own-price effect: $\partial x_1 / \partial p_1 = -b / (p_1 + a \cdot p_2)^2 < 0$ (always negative). Demand falls as its own price rises — the standard downward-sloping case — so good 1 is an ordinary good, not a Giffen good.

Problem 6

Human capital refers to the stock of skills, knowledge, health, training, and experience embodied in a person that makes their labor more productive. Economists treat it as a form of capital because, like a machine or a building, it is built up through deliberate investment — education, on-the-job training, healthcare — that carries an upfront cost but pays off over time through higher productivity and higher earnings. Unlike physical capital, however, human capital cannot be bought, sold, or separated from the individual who possesses it; it is embodied in the person.

Other major types of capital recognized in economics:

- Physical (or produced) capital — machinery, tools, buildings, and equipment used to produce goods and services.
- Financial capital — money, stocks, bonds, and other funds used to acquire the other forms of capital.
- Natural capital — land, raw materials, and natural resources/ecosystems that provide productive inputs.
- Social capital — networks, relationships, trust, and shared norms that make cooperation and economic exchange easier and less costly.

Problem 7

Landsburg's point: a country can "produce" a good either by making it directly, or by producing something else and trading for it — and from the standpoint of the economy, those two routes are economically interchangeable. Applied to US steel and soybeans:

- Step 1: The US has land, labor, and capital that can be used to grow soybeans directly — call this the "direct" technology for producing soybeans.

- Step 2: Alternatively, those same resources can be used to produce steel.
- Step 3: The US then trades that steel abroad (e.g., to a country with a comparative advantage in growing soybeans) in exchange for soybeans.
- Step 4: From a national accounting standpoint, the input was US labor/capital (used to make steel) and the output is soybeans available for US consumption — functionally identical to a technology that transforms steel into soybeans.
- Step 5: Whether the US grows soybeans on a farm or "grows" them by exporting steel and importing soybeans in return, the end result is the same bushels of soybeans on the market. International trade is therefore just an alternative production technology, and comparative advantage determines which "technology" (direct farming vs. trade-via-steel) is more efficient.

Tariff on Chinese steel as a tax on soybeans: a tariff on imported steel raises the price of steel in the US, which makes the "steel-for-soybeans" trade technology less efficient — it now takes more resources (or yields fewer soybeans) to obtain the same quantity of soybeans via this indirect route, exactly as if a tax had been imposed directly on the technology that produces soybeans. There is also a balance-of-trade channel: if the US imports less Chinese steel, China earns fewer dollars and/or the exchange rate adjusts, which — all else equal — makes it harder for the US to sell as many soybeans to China (and China may respond with retaliatory tariffs on US soybeans directly, as happened in practice). Either way, protecting the domestic steel industry with a tariff acts as an implicit tax on the soybean-exporting sector. This is a version of the Lerner symmetry result: a tax on imports is economically equivalent to a tax on exports, because trade must balance overall.